Xinglang Zhang

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Education

Huazhong University of Science and Technology (HUST) Sept 2024 – Jun 2028

B.S. in Cyber Science and Engineering

- **GPA:** 4.80/5.0 **Weighted Avg:** 94.51 **Rank:** 1/108
- **Selected Coursework:** Calculus A (I/II) (A+), Linear Algebra (A+), Discrete Math (A+), Data Structures (A+), Assembly Language (A+), Probability Theory (A+), Digital Circuit (A+), Database Systems (A+).

Research Agenda

Reliable LLM Reasoning. When and why does reasoning break down, and how can structured or neuro-symbolic mechanisms make it more reliable?

Test-Time Learning & Self-Evolving Agents. How can agents evaluate the value of self-generated experiences and decide what to learn, retain, and reuse as their behavior evolves?

Human–AI Preference & Social Intelligence. Where does model behavior diverge from human preferences and social signals, and how can these gaps inform model design?

Research Experience

Understanding Multi-Agent Systems Nov 2026 – May 2027

Research Intern, System Intelligence Group, Microsoft Research Asia Beijing, China

Research Focus: Multi-Agent Systems

Adaptive Learning & Self-Evolving Agents

TRACE: Transfer-Aware Test-Time Training for Tool-Using LLMs Jul 2026 – Present

Research Intern, BLENDER Lab, UIUC

Mentor: Cheng Qian **Lab Director:** Heng Ji

- **Question.** Test-time training can learn from self-generated tool-use practices, but a limited update budget makes experience selection essential. Formulated the question: *which experience is worth learning now, given what the model has already learned?*
- **Finding.** Characterized *Transfer Value* as the benefit of learning from an experience for a target distribution. Established that this utility is **relational and state-dependent**: the same experience can help one query and harm another, and its value can reverse as the solver updates. Demonstrated cross-query transfer and non-monotonic changes in utility. In a controlled probe, **30 of 32** initially beneficial experiences became harmful after 32 updates; some became useful again later, motivating repeated selection rather than a fixed ranking.
- **Method.** Developed *TRACE* to combine *Target Relevance*, *Current-State Transferability*, and *Marginal Complementarity*. Temporary updates and gold-free probes allocate the learning budget; complementary practices are selected, learned through persistent LoRA-SFT, and re-evaluated as the solver evolves.
- **Evidence.** Under matched permanent update budgets, probing the updated solver reached **38.1 Exact** versus **30.4** with stale-state probes. Reordering the same selected experiences degraded performance, showing that both *what* and *when* to learn matter. Across four tool-use benchmarks and four main models, improved Overall Exact accuracy by **13.10–24.42 percentage points** over frozen models and consistently outperformed static selection. These findings frame effective TTT as **dynamic curriculum formation**. First-author ACL 2027 submission, under review [3].

Anchor: Locally Verified Memory for Self-Evolving Agents Aug 2026 – Oct 2026

Research Collaborator, HKU CDS

Advisor: Jian Qian

- **Question.** Co-formulated the problem of experience valuation in self-evolving agents: self-generated memories can be plausible yet redundant or harmful, while trajectory-level feedback does not reveal their actual decision value. Asked whether a memory's influence on its source decision can provide evidence for selective retention.
- **Method.** Co-proposed Anchor, a memory-admission framework that replays source decisions with and without candidate memories under a frozen policy. Combined **behavioral shift magnitude** (Jensen–Shannon divergence) with **source-role alignment**, retaining memories that reinforce successful actions or correct failed ones before incorporating them into an evolving memory bank.
- **Evidence.** Co-conducted experiments on ALFWorld and WebShop across two LLM backbones. With Qwen3.5-27B, Anchor achieved **96.27% ALFWorld success** and **73.50 WebShop mean score**, outperforming the strongest respective baselines (92.54% and 62.74), while retaining **41% fewer memories** than unfiltered storage. Second-author paper [9].

Research Experience (continued)

Reliable LLM Reasoning

Logical Phase Transitions: Diagnosing and Mitigating Reasoning Collapse Aug 2025 – Jan 2026
Project Leader, Intelligent Media Computing and Network Security Lab, HUST
Advisor: Zikai Song

- Investigated where reasoning becomes unreliable as *logical complexity* increases, asking whether collapse follows identifiable regimes that aggregate benchmark scores obscure.
- **LoCM and NSA-LR.** Introduced the *Logical Complexity Metric (LoCM)* and *NSA-LR*, pairing natural-language statements with explicit first-order-logic representations and reasoning chains. Enabled complexity analysis beyond coarse hop-count estimates.
- Identified *Logical Phase Transitions*: accuracy remains relatively stable within complexity regimes, then drops sharply across critical intervals. Observed this pattern across open- and closed-source LLMs.
- Developed *Neuro-Symbolic Curriculum Tuning*, combining natural-language/symbolic alignment with complexity-aware training around the observed transition regions to progressively strengthen reasoning at higher complexity.
- Across five benchmarks, obtained average accuracy gains of **+1.26** with naive prompting and **+3.95** with CoT, alongside improved generalization to unseen logical compositions. Connected failure diagnosis to curriculum design. Co-first-author paper, **ACL 2026 Main** [1].

From LogicAgent to HexLogicAgent: Structuring Meaning Before Deduction
Research Assistant, Intelligent Media Computing and Network Security Lab, HUST
Advisor: Zikai Song
LogicAgent: Feb – Aug 2025 HexLogicAgent: Mar – Jul 2026

- Investigated how *semantic complexity* interacts with *logical depth*: valid deduction depends on first interpreting abstract, ambiguous, or conflicting propositions correctly.
- **LogicAgent.** Contributed to a semiotic-square-guided framework that structures alternative meanings before automated deduction and reflective verification. Introduced *RepublicQA* to jointly evaluate semantic and logical difficulty; achieved **+6.25%** average improvement on RepublicQA and **+7.05%** across four mainstream benchmarks. Co-authored **ACL 2026 Main / Oral** paper [2].
- The square leaves mixed and partially compatible meanings underrepresented. Identified incomplete semantic representations as a major source of failures despite apparently valid deduction.
- **HexLogicAgent.** Extended the semantic structure from a square to a logical hexagon, explicitly representing uniform and mixed cases before deduction and verification. Developed a formal account of semantic opposition and completeness under stated first-order-logic assumptions.
- HexLogicAgent improved reliability across five benchmarks and delayed degradation as logical complexity increased. Established **semantic organization before inference** as a focus of this research line. Co-first-author *Artificial Intelligence* submission [5].

Human–AI Preference & Social Intelligence

OMRA: Diagnosing and Reducing Synthetic–Real Preference Misalignment Jan – May 2026
Project Leader, Intelligent Media Computing and Network Security Lab, HUST
Advisor: Zikai Song

- Asked whether the qualities LLMs associate with high engagement match real user responses. Studied **1.17 million answers** to **25,978 questions** on Zhihu, Quora, and Reddit; introduced *Ontological Preference Measurement* along logic, affect, and expression dimensions.
- Identified *logic overbinding*: as target engagement increases, LLMs add explicit reasoning structure, while observed user engagement is more strongly associated with affective and expressive salience.
- Developed *Ontology-Masked Reasoning Autoencoding (OMRA)* to mask and reconstruct over-explained spans while preserving stance, factual content, and coherence.
- Reduced marginal, geometric, and transfer gaps by **54.4% on average** across four LLM families. The results motivate grounding preference alignment in observed social signals. First-author NAACL 2027 submission [4]; all three reviewers gave **OA 3.5/5**, and the Area Chair gave a **meta-review score of 4**, with a reported top-8% ranking.

Related collaborative work.

- Co-authored *IntervenSim* [7], modeling the co-evolution of interventions and crowd opinion; co-authored the *Social Intelligence Modeling* survey [6], connecting social perception with social simulation.

Publications

* Equal contribution.

[1] Logical Phase Transitions: Understanding Collapse in LLM Logical Reasoning
Xinglang Zhang*, Yunyao Zhang*, et al.* | **ACL 2026 (Main; co-first author)** | [ArXiv] | [Code]

[2] Semantic-Aware Logical Reasoning via a Semiotic Framework
*Yunyao Zhang, **Xinglang Zhang**, et al.* | **ACL 2026 (Main; Oral)** | [ArXiv] | [Code]

[3] TRACE: Transfer-Aware Test-Time Training for Tool-Using LLMs
***Xinglang Zhang**, Cheng Qian* | [Paper] | **ACL 2027, under review**

[4] Too Good to Be Real? Diagnosing and Reducing the Gap Between AI Preference and Real User Engagement
***Xinglang Zhang**, Yuanmeng Xiang, et al.* | [ArXiv] | **NAACL 2027 submission**

[5] Semiotic Logical Hexagon Theory for LLM Logical Reasoning
Yunyao Zhang, **Xinglang Zhang***, et al.* | [ArXiv] | [Code] | **Co-first author; Artificial Intelligence submission**

[6] Social Intelligence Modeling: A Comprehensive Survey from Social Perception to Social Simulation
*Zikai Song, Xiajie Li, Yunyao Zhang, **Xinglang Zhang**, et al.* | [Paper] | [GitHub]

[7] IntervenSim: Intervention-Aware Social Network Simulation for Opinion Dynamics
*Yunyao Zhang, Zuocheng Ying, **Xinglang Zhang*** | [ArXiv] | [Code]

[8] EarthVerse: Benchmarking Scientific Agents Across Dynamic Earth Systems and Natural Hazards
*Zhiqing Cui, Xinxiang Yin, Yihong Tang, **Xinglang Zhang**, et al.* | [Project Page] | [Paper]

[9] From Reflection to Co-Evolution: Self-Evolving Agents with Locally Verified Memory
*Xinxiang Yin, **Xinglang Zhang**, et al.* | [Paper]

Selected Funded Projects

- **HUST Undergraduate Natural Science Fund, 2026**
  Trustworthy Enhancement for Complex Reasoning
  Project Lead | **CNY 50,000 (approx. US$7,450)**
- **College Student Innovation and Entrepreneurship Program (University Level)**
  LLM Logical Reasoning through Philosophical Paradigms
  Project Lead | **CNY 4,000 (approx. US$600)**
- **College Student Innovation and Entrepreneurship Program (Hubei Provincial Level)**
  Trustworthy Enhancement of LLMs via a Neuro-Symbolic Framework
  Project Lead | **CNY 20,000 (approx. US$2,980)**

Honors

- **Outstanding Student, Qiming College** *(Highest Honor for HUST Undergraduates; Top 1.85%)* 2025 – 2027
- **Jin Yugang Lan-Ying Scholarship** *(Top 1%)* 2025 – 2026
- **National Scholarship** *(Highest Honor for Chinese Undergraduates; Top 0.3% Nationwide)* 2025 – 2026
- **Triple-A ("Sanhao") Student Scholarship** *(Top 5.5%)* 2025 – 2026
- **National Scholarship** *(Highest Honor for Chinese Undergraduates; Top 0.3% Nationwide)* 2024 – 2025
- **Triple-A ("Sanhao") Student Scholarship** *(Top 5.5%)* 2024 – 2025

Service

Reviewer, ACL ARR January, May, and August 2026

Academic Activities

ACL 2026 Poster Presenter and Oral Presenter July 1 – 8, 2026
Selected Rising Star Poster Presenter, X-AGI Conference Oct. 17 – 18, 2026

Skills

ML & Research Stack: Python, C/C++, PyTorch, HuggingFace, LoRA, SFT, RLHF, vLLM

Mathematical Foundations: Linear Algebra, Probability Theory, Calculus, Discrete Mathematics

Languages: English (TOEFL 5.5/6; W6 R6 L5.5 S5; CET-4: 609; CET-6: 611), Chinese (Native)

Leadership Experience

- **Class Representative, Qiming Experimental Class 2401** Sept 2024 – Present
  - Organized academic events, managed administrative tasks, and mentored students to enhance class cohesion.
- **Head of Organization Department, School of Cyber Science and Engineering** Sept 2025 – Present
  - Managed administrative and organizational affairs for the Communist Youth League.
- **Head of Lesson Planning, Peer Lecturer Group** Apr 2025 – Present
  - Fostered academic excellence across the college through peer mentoring and curriculum planning.